

7 The Zeta Function and Prime Number Theorem

Bernhard Riemann, whose extraordinary intuitive powers we have already mentioned, has especially renovated our knowledge of the distribution of prime numbers, also one of the most mysterious questions in mathematics. He has taught us to deduce results in that line from considerations borrowed from the integral calculus: more precisely, from the study of a certain quantity, a function of a variable s which may assume not only real, but also imaginary values. He proved some important properties of that function, but enunciated two or three as important ones without giving the proof. At the death of Riemann, a note was found among his papers, saying “These properties of $\zeta(s)$ (the function in question) are deduced from an expression of it which, however, I did not succeed in simplifying enough to publish it.”

We still have not the slightest idea of what the expression could be. As to the properties he simply enunciated, some thirty years elapsed before I was able to prove all of them but one. The question concerning that last one remains unsolved as yet, though, by an immense labor pursued throughout this last half century, some highly interesting discoveries in that direction have been achieved. It seems more and more probable, but still not at all certain, that the “Riemann hypothesis” is true.

J. Hadamard, 1945

Euler found, through his product formula for the zeta function, a deep connection between analytical methods and arithmetic properties of numbers, in particular primes. An easy consequence of Euler’s formula is that the sum of the reciprocals of all primes, $\sum_p 1/p$, diverges, a result that quantifies the fact that there are infinitely many prime numbers. The natural problem then becomes that of understanding how these primes are distributed. With this in mind, we consider the

following function:

$$\pi(x) = \text{number of primes less than or equal to } x.$$

The erratic growth of the function $\pi(x)$ gives little hope of finding a simple formula for it. Instead, one is led to study the asymptotic behavior of $\pi(x)$ as x becomes large. About 60 years after Euler's discovery, Legendre and Gauss observed after numerous calculations that it was likely that

$$(1) \quad \pi(x) \sim \frac{x}{\log x} \quad \text{as } x \rightarrow \infty.$$

(The asymptotic relation $f(x) \sim g(x)$ as $x \rightarrow \infty$ means that $f(x)/g(x) \rightarrow 1$ as $x \rightarrow \infty$.) Another 60 years later, shortly before Riemann's work, Tchebychev proved by elementary methods (and in particular, without the zeta function) the weaker result that

$$(2) \quad \pi(x) \approx \frac{x}{\log x} \quad \text{as } x \rightarrow \infty.$$

Here, by definition, the symbol $\approx$ means that there are positive constants $A < B$ such that

$$A \frac{x}{\log x} \leq \pi(x) \leq B \frac{x}{\log x}$$

for all sufficiently large x .

In 1896, about 40 years after Tchebychev's result, Hadamard and de la Vallée Poussin gave a proof of the validity of the relation (1). Their result is known as the prime number theorem. The original proofs of this theorem, as well as the one we give below, use complex analysis. We should remark that since then other proofs have been found, some depending on complex analysis, and others more elementary in nature.

At the heart of the proof of the prime number theorem that we give below lies the fact that $\zeta(s)$ does not vanish on the line $\text{Re}(s) = 1$. In fact, it can be shown that these two propositions are equivalent.

1 Zeros of the zeta function

We have seen in Theorem 1.10, Chapter 8 in Book I, Euler's identity, which states that for $\text{Re}(s) > 1$ the zeta function can be expressed as an infinite product

$$\zeta(s) = \prod_p \frac{1}{1 - p^{-s}}.$$